



Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU

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LANDSLIDE AND FLOOD THREAT ADVISORIES

**To: Brgy. Chairperson
Brgy. Nangka
City of Marikina
Metropolitan Manila**

Date: Oct 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **landslide and flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Barangay Hall	N 14° 40' 30.9" E 121° 06' 26.6"	High	Nangka is identified as one of the most flood susceptible barangays in Marikina. This is primarily because it is bounded on its west by the Marikina River and on the north-east by the Nangka River. The confluence of the two channels is located just north of the barangay. High influx of rainfall and runoff causes the channels to overflow and inundate the nearby areas.
Nangka River	N 14° 40' 31.1" E 121° 06' 26.8"	Very High	
Nangka River – Marikina River confluence	N 14° 40' 36.5" E 121° 06' 19.3"	Very High	
Bayabas – Carancho Boundary	N 14° 40' 30.3" E 121° 06' 13.6"	Very High	The surrounding area of the confluence of Nangka River and Marikina River experienced flood heights exceeding the roof of one-storey houses during Ulysses. A section of the riverwall/dike along the San Mateo side of Nangka River was damaged recently (May 2023). The residential area of Bayabas Ext., situated behind the dike of Marikina River, experienced flood heights >2m.
Balubad-Bayabas	N 14° 40' 29.2" E 121° 06' 15.3"		The vicinity of the barangay hall is proximal to Nangka River, which acts as the boundary with San Mateo, Rizal. The house in front of the hall (interior is lower than the road), the flood height during Ondoy reached around 1.3m, while only the flooring of the hall was inundated. However, during Ulysses, no flooding was reported in the area.
Kabayani	N 14° 40' 25.06" E 121° 06' 11.9"		Meanwhile, Kabayani Rd. runs parallel to Marikina River and proximal to it. Flooding during Ondoy exceeded 2m as reported by multiple households. Even greater heights were experienced, during the same event, in the Settlement, where floodwaters reached the roofs of two-storey houses, as noted by a resident.
	N 14° 40' 23.2" E 121° 06' 11.3"	High	
Eskenita to Settlement	N 14° 40' 16.9"		

Settlement	E 121° 06' 04"	Very High	Several subdivisions or residential communities are flooded at times of heavy rainfall. In Marikina Village, Sierra Madre is slightly elevated and only experience gutter-deep flooding, but other portions of the subdivision were submerged by greater than 1m-high floodwaters during Ondoy. In Terra Vista Subdivision, despite being proximal to Nangka River, the subdivision is relatively elevated, with low-lying portions near Anastacia Village. These areas reported maximum flood heights ~1m.
	N 14° 40' 16.9" E 121° 06' 01.1"		
F. Manalo	N 14° 40' 15.5" E 121° 06' 58.8"	High	On the other hand, compared to surrounding subdivisions, Anastacia Village is relatively depressed with flood heights greater than 2m during Ondoy, reaching the ceiling/roof of the 1 st floor.
	N 14° 40' 25.1" E 121° 06' 17.5"		
Nangka Elementary School	N 14° 40' 25.7" E 121° 06' 18.7"	High	Relative to Kabayani Rd., F Manalo St. is slightly more elevated and is bounded along its eastern side by the wall of Greenland Subdivision. Flooding along this street does not exceed 0.5m. As it reaches the vicinity of the Elementary School, flood heights recorded ~1m during Ondoy, while none were reported in Ulysses.
Nangka Health Center	N 14° 40' 26.9" E 121° 06' 27.6"		
House near Nangka River	N 14° 40' 27.4" E 121° 06' 29.5"	Very High	
	N 14° 40' 26.9" E 121° 06' 21.8"		

Remarks / Recommendations:

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.
- ✓ **Construction and/or improvement of drainage system.** Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes

- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ **Construction of appropriate and well-studied engineering measures** (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.

Prepared by:


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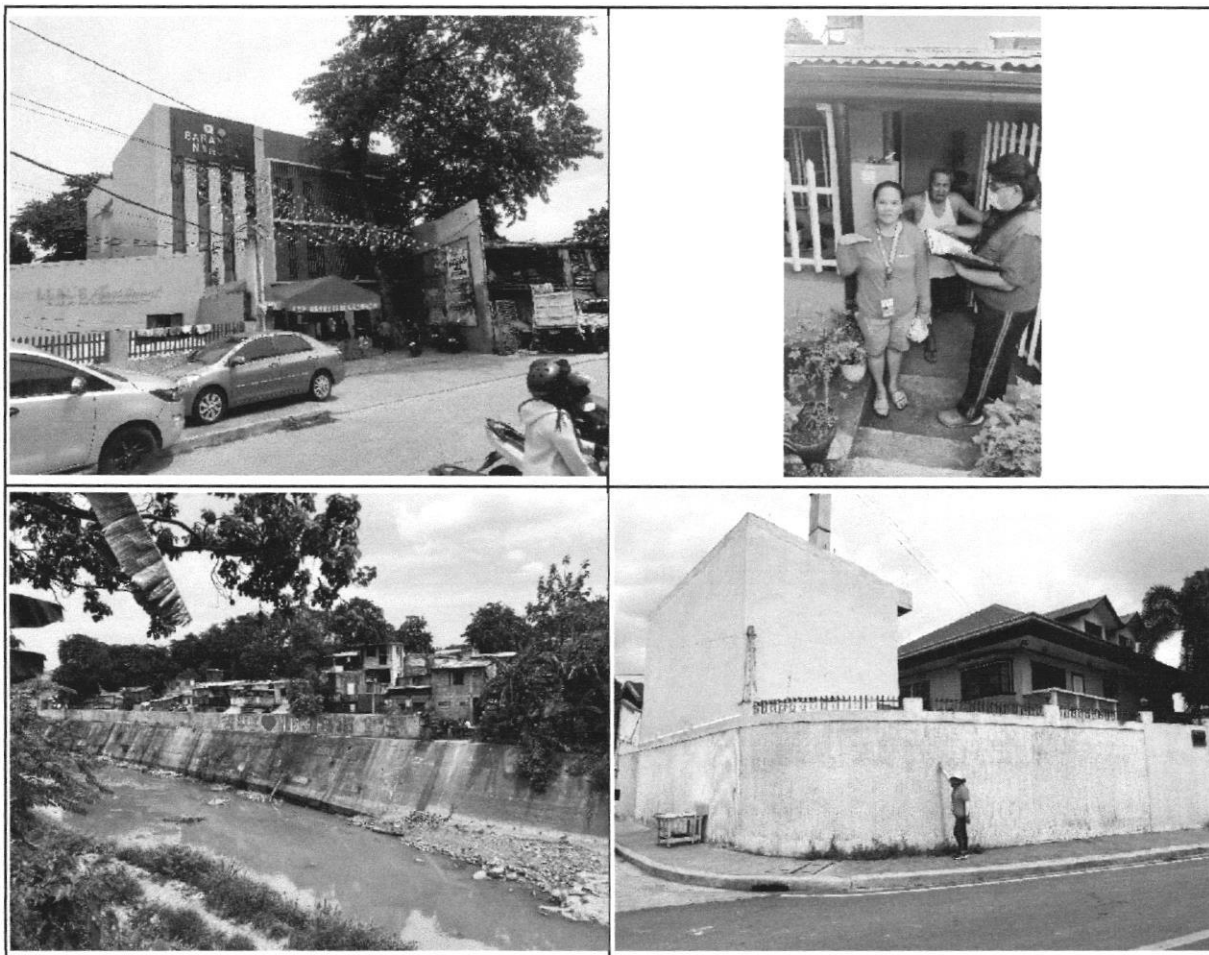

JETHRO B. CAPINO
Geologist II

Received by:

Designation:

Office:

PHOTOS

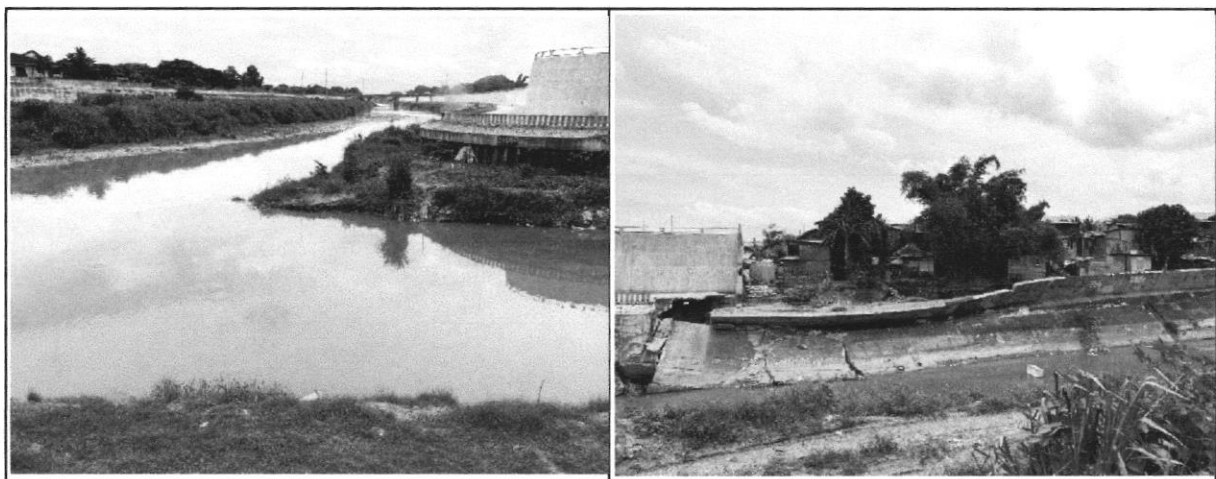


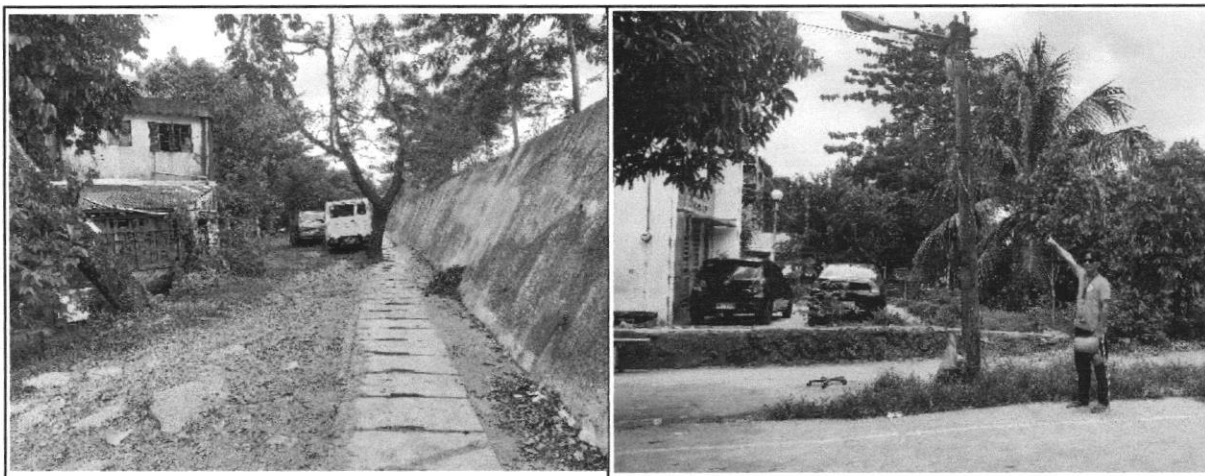
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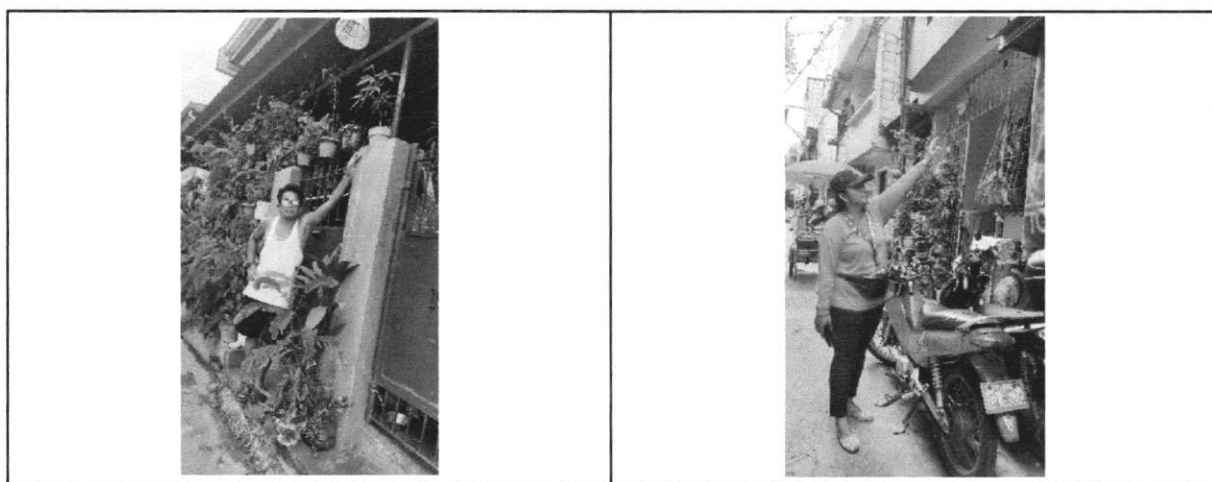
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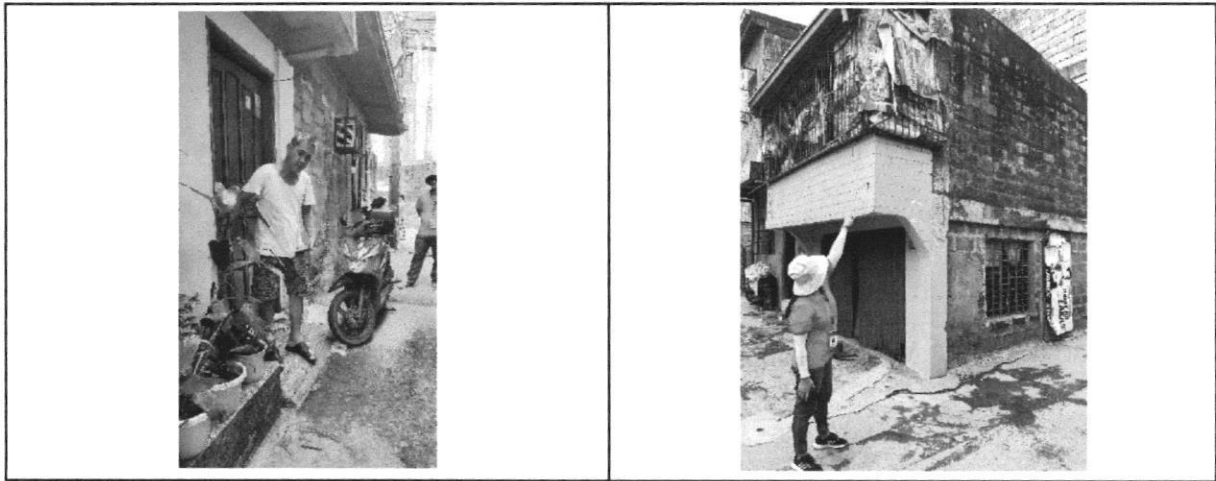
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Several subdivisions or residential communities are flooded at times of heavy rainfall. Photos show the flood heights in different areas of the barangay.

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